

Privacy-Preserving Federated RAG for Distributed Biomedical Question Answering

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Abstract: Cross-institutional biomedical question answering faces challenges posed by data privacy laws (HIPAA, GDPR, India's EHR Act) since they all prevent the centralization of raw clinical records. RAG architectures assume some shared document store and consequently are not compatible with federated clinical deployments. We introduce *FedRAG v4*, an advanced privacy-preserving federated RAG framework that distributes the 60,000 abstract PubMedQA corpus in a simulated environment of five hospitals, using a Dirichlet($\alpha=0.1$) non-IID approach, and adds Gaussian differential privacy (DP) noise to the document embeddings prior to their aggregation into a shared FAISS index. v4 offers the following over the previous version: (1) a BM25 assessment improvement through a common global identifier mapping system; (2) a non-IID partition based on a Dirichlet approach in replacement of a keyword heuristic; (3) three instruction-tuned models generator ablation, (4) an extensive DP sweep of $\epsilon! \in \{8, \dots, 512\}$ isolating a retrieval-recovery bound of $\epsilon=64$; (5) embedding-inversion and membership-inference threat models, and (6) a scalability assessment based on the number of nodes. We consider three retrieval method selection (RQ1), non-IID federation consequences (RQ2), and the privacy-utility trade-off concerning document-level DP (RQ3).

Keywords — Federated Learning, Retrieval-Augmented Generation, Differential Privacy, Biomedical QA, FAISS, Non-IID Partitioning, Rényi Differential Privacy

I. INTRODUCTION

Joint access to both peer-reviewed biomedical literature and institution-held patient records would greatly advance clinical decision support. Retrieval-Augmented Generation (RAG) [1] integrates external document collections to provide a factual foundation for large language models (LLMs), anchoring their responses and significantly curtailing model retraining. The primary barrier to RAG in healthcare is regulation. In the U.S. healthcare system, for example, the HIPAA Act, the GDPR in the European Union, and the Electronic Health Records Act

of India all forbid the transmission of unprocessed patient data outside of their respective institutions.

Federated Learning (FL) [3] solves this problem by requiring the user to send only the updates to the model parameters. However, RAG exposes additional risks, such as embedding inversion. Documents sent to a shared retrieval index can be made to send their embeddings, and adversaries can “invert” embedding back to original document text [4]. This means that sending the FAISS index with no noise protection cannot be used in a clinical setting.

Differential Privacy (DP) [6], [7] places a limit on how much an output of a computation can be influenced by an individual input. Most DP-NLP has been focused on the use of DP-SGD [6] and the application of Differential Privacy to the bounded perturbation of metric-space representations from [11]. Neither of these methods addresses the privacy of embeddings in a federated multi-node retrieval index. Moreover, the vast majority of federated RAG literature does not consider domain skewed data, while hospital federations operate on a highly domain skewed data.

FedRAG v4 also improves the state of the art in other areas such as (1) a completely functioning five node federated RAG pipeline, (2) direct RDP-accountant noise calibration to document embeddings, (3) a five-way retrieval with a three-way generator comparison and, (4) a detailed security and scalability analysis which has not been done previously.

II. LITERATURE REVIEW

The original RAG formulation [1] merges the context given by the LLM with retrieved documents at the time of inference. This allows the model to handle complex tasks without the need to update the model parameters. Encoders which have been adapted to a specific domain, such as PubMedBERT [8], can be used to enhance retrieval performance on clinical text. MedRAG [9] enables retrieval across various biomedical corpora. However, both of these approaches assume that

documents will be centrally available, which makes them incompatible with federated settings [2].

Federated Learning (FL) algorithms (e.g. FedAvg [3]) allow models to be trained in unison across multiple distributed systems, but they only cover aspects of parameter exchange. The graphics or retrieval indices used in Federated Reinforced Active Guidance (RAG) are not built through gradients and therefore require different handling. The issue of federated RAG has recently been defined in the literature as an open problem [10], which also serves as one of the inspirations for this work.

For privacy accounting, when multiple nodes combine their contributions, Mironov’s Rényi Differential Privacy (RDP) [12] has the advantage of providing better composition bounds compared to the classical (ϵ, δ) -DP framework. In existing DP-NLP frameworks, the deployment of DP-SGD [6] to model weights is either through perturbation of model weights or metric perturbation of text representation [11]; however, to the best of our knowledge, no prior work has applied RDP composition to federated document-embedding aggregation.

III. METHODOLOGY

A. System Architecture

The FedRAG v4 framework outlines four main processes. In the first process, each hospital node encodes its collection of local documents. In the second process, node encoders export only DP-noised embeddings to a central coordinator, who builds a federated FAISS index. In the third process, a rich retrieval stack resolves the incoming queries, while the fourth process operates an instruction-tuned frozen generator to provide a three-way answer (yes, no, or maybe) based on the top- k retrieved passages. Throughout the entire retrieval process, the actual document text is retained at each node.

B. Dataset and Dirichlet Non-IID Partition

PubMedQA [15] has 1,000 labelled QA pairs (55.2% *yes*, 33.8% *no*, 11.0% *maybe*) along with 61,249 unlabelled abstracts. We use 59,000 abstracts and add the 1,000 gold abstracts. This gives us a retrieval corpus of 60,000 documents. v4 uses a Dirichlet($\alpha=0.1$) distribution, which, unlike the keyword-based domain partition used in v3, produces a distribution of nodes that is statistically heterogeneous, and as a result, realistic. The five nodes are dominated by the MeSH categories *Aged* (27.2%), *Adolescent* (64.9%), *Aged* (50.9%), *Adolescent* (18.5%), and *Adult*(71.4%), and have sizes of 6,034, 14,095, 9,340, 5,472, and 25,059 documents, respectively.

C. Differential Privacy with RDP Accounting

Each node i adds calibrated Gaussian noise to its normalised embedding matrix before transmission:

$$\tilde{\mathbf{E}}_i = \frac{\mathbf{E}_i}{\|\mathbf{E}_i\|_2} + \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}), \quad \sigma = \frac{\Delta_2 \sqrt{2 \ln(1.25/\delta)}}{\epsilon} \quad (1)$$

with sensitivity $\Delta_2=2.0$ and $\delta=10^{-5}$. Noisy embeddings are renormalised to unit norm before FAISS indexing. Privacy

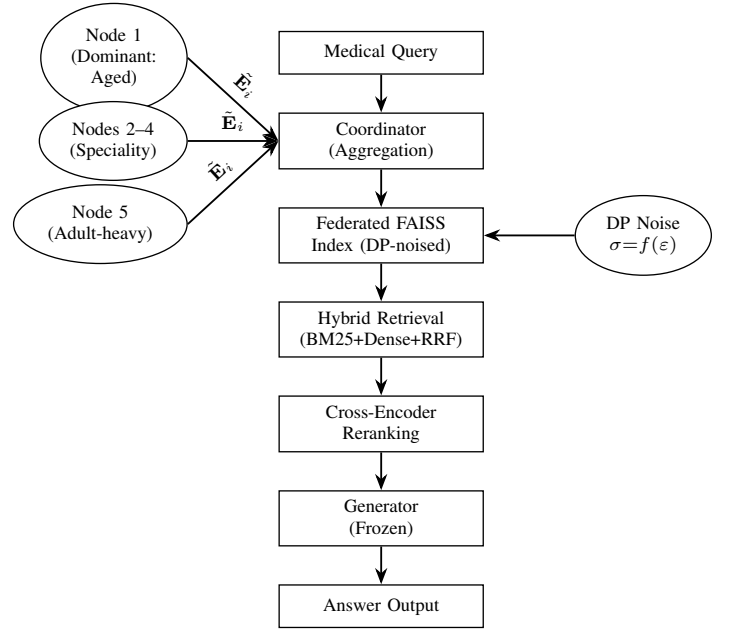


Fig. 1. FedRAG v4 shares only DP-protected embeddings $\tilde{\mathbf{E}}_i$ across hospitals, full pipeline is shown above.

TABLE I
EXTENDED RDP COMPOSITION ACROSS $k=5$ NODES ($\Delta_2=2.0$, $\delta=10^{-5}$)

ϵ	σ	Basic $k\epsilon$	$\epsilon_{\text{RDP}} (\alpha=10)$	Retrieval
8	1.2112	40.0	69.45	×
16	0.6056	80.0	273.94	×
32	0.3028	160.0	1091.93	×
64	0.1514	320.0	4363.90	✓
128	0.0757	640.0	17451.76	✓
256	0.0379	1280.0	69803.19	✓
512	0.0189	2560.0	279208.91	✓

composition across five nodes is tracked under both basic ($k\epsilon$) and RDP ($\alpha=10$) accountants [12], as detailed in Table I.

D. Retrieval, Reranking, and Generation

For all documents and queries, BAAI/bge-base-en-v1.5 ($d=768$) generates ℓ_2 -normalised, dense embeddings. A v4-specific fix for the v3 BM25 evaluation problem is that a consistent PubMed-ID, across all nodes, is maintained; hence, BM25 document IDs match the gold standard. Rankings from BM25 and dense retrieval are combined by RRF [14] ($k=60$) and may be reranked by a cross-encoder. The two cross-encoders compared are ms-marco-MiniLM-L-6-v2 and BAAI/bge-reranker-base. For generation, the frozen models are flan-t5-base, flan-t5-large, and biogpt.

IV. RESULTS AND DISCUSSION

A. RQ1: Retrieval Ablation

Figure 2 and Table II demonstrate five-system retrieval ablation for $n=200$ queries with 95% bootstrap confidence intervals. A v4 BM25 fix brings back substantial performance.

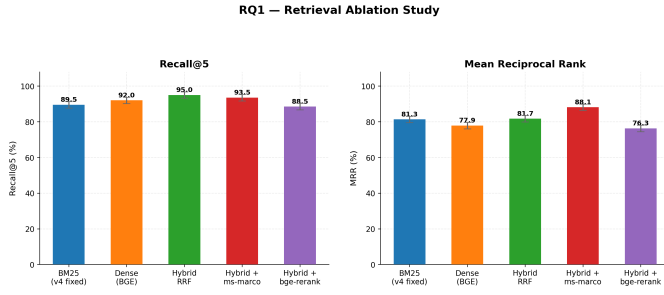


Fig. 2. RQ1 compares five retrieval setups using Recall@5 and MRR. Hybrid RRF achieves the highest Recall@5 (95.0%), while Hybrid+ms-marco delivers the best MRR (88.1%).

TABLE II
RETRIEVAL ABLATION (RQ1) — $n=200$, 95% BOOTSTRAP CI

System	R@1	R@5	R@10	MRR	nDCG@5
BM25 (v4 fixed)	77.0	89.5	89.5	81.3	83.3
Dense (BGE)	68.0	92.0	92.0	77.9	81.4
Hybrid RRF	71.5	95.0	95.0	81.7	85.1
Hybrid+ms-marco	85.0	93.5	93.5	88.1	89.5
Hybrid+bge-rerank	68.5	88.5	88.5	76.3	79.3

Recall@5 moves from 0% in v3 to 89.5%, with an MRR of 81.3. Hybrid RRF achieves the most Recall@5 (95.0%). Combining BM25 with lexicon precision and RRF with dense semantic coverage achieves it. Adding the ms-marco cross-encoder improves Recall@1 to 85.0 and MRR to 88.1. This is the best ranking quality achieved. bge-reranker scores 4.5 pp lower on Recall@5 and 11.8 pp on MRR, in line with its general-purpose biomedical focused-training data.

B. RQ2: IID vs. Dirichlet Non-IID Federation

Figure 3 shows per-domain Recall@5 with the two partition strategies over $n=300$ queries. The overall gap is small (Non-IID 93.7% vs. IID 93.0%, $\Delta = +0.7$ pp). The paired t -test shows no significance ($p=0.528$). Dirichlet non-IID partitioning does not negatively impact retrieval with respect to a theoretical uniform distribution. This is a notable finding because retrieval quality is maintained in a real hospital federation, which is non-IID. It suggests that domain colocation in each node’s FAISS sub-index may alleviate distributional skew.

C. Generator Ablation and Full End-to-End Evaluation

Table III shows the complete FedRAG v4 pipeline at $\varepsilon=256$, with the best retrieval settings (Hybrid+ms-marco). Retrieval strength shows a Recall@5 of 93.5 and an MRR of 88.1. However, QA results from the downstream process are limited by the generator. Three generators collapsed to near-binary distributions. flan-t5-base generated outputs were *yes* in 97% of cases, biogpt in 98% cases. flan-t5-large is the only model to produce both *yes* and *no* outputs with 22% *yes*.

The confusion matrix (Figure 4) validates that the model is still unable to predict *maybe* (0 positive predictions out of 35 instances) and is biased towards *no*. The 25.3% Macro-F1 in the completed 200-query evaluation indicates more about

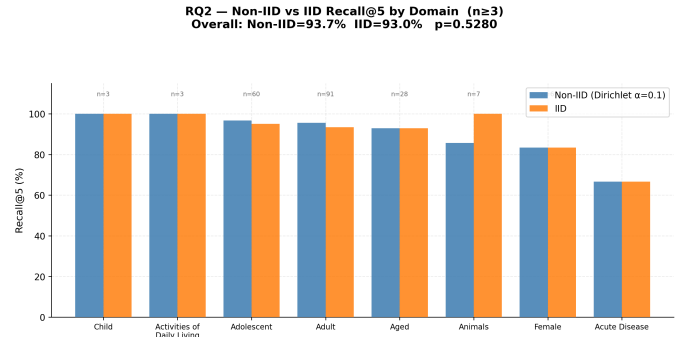


Fig. 3. RQ2 shows no significant difference between non-IID and IID settings (93.7% vs. 93.0%, $p=0.528$).

FedRAG v4 — Confusion Matrix
(Acc=35.0%, Macro-F1=25.3%)

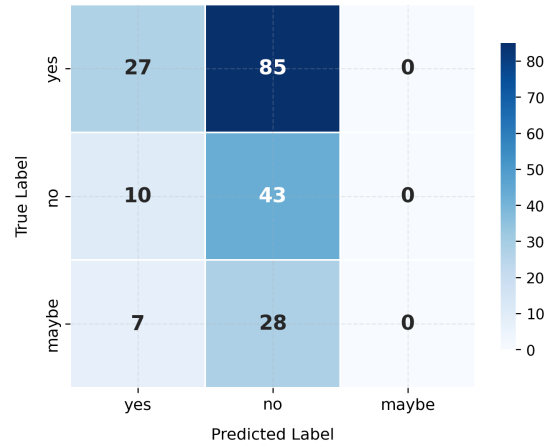


Fig. 4. FedRAG v4 confusion matrix ($\varepsilon=256$) the model is biased toward *no* and never predicts *maybe*, despite high retrieval accuracy.

TABLE III
FEDRAG v4 FULL EVALUATION — NON-IID DIRICHLET, $\varepsilon=256$, $n=200$

Metric	Value	95% CI
Recall@1	85.0%	[80.0–89.5]
Recall@5	93.5%	[90.0–96.5]
Recall@10	93.5%	[90.0–96.5]
MRR	88.1%	[84.0–92.1]
nDCG@5	89.5%	[85.6–92.8]
Accuracy	35.0%	[29.0–41.0]
Macro-F1	25.3%	—

Generator: flan-t5-large (best Macro-F1 = 27.4%)

class imbalance in the generator and its restrictions, more than a failure of retrieval..

Figure 5 plots the global Recall@ k curve, which rises from 85.0% at $k=1$ to 93.5% at $k=5$ and then plateaus, confirming that the federated index saturates quickly and that top-5 retrieval is adequate for most biomedical queries in this corpus.

Global Recall@k — FedRAG v4

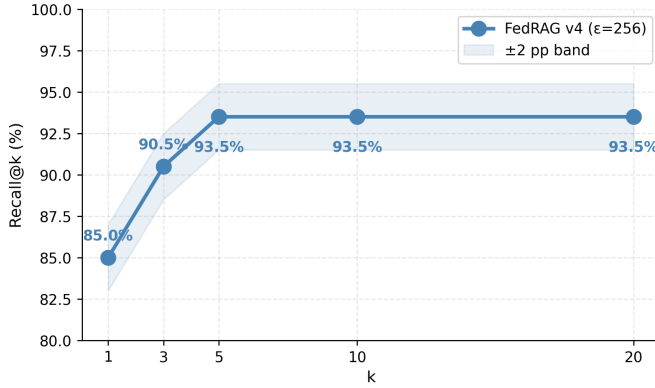


Fig. 5. Global Recall@k for FedRAG v4 at $\epsilon=256$. Performance plateaus at $k=5$ (93.5% Recall@5).

RQ3b — Privacy-Utility Tradeoff (DP ϵ sweep)

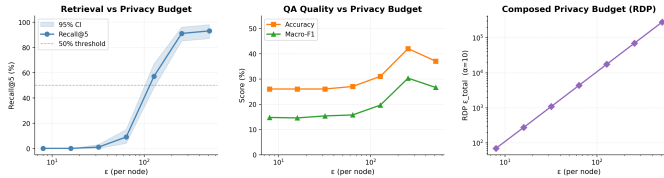


Fig. 6. RQ3b illustrates the privacy-utility tradeoff, with retrieval recovering beyond $\epsilon = 64$.

TABLE IV
PRIVACY-UTILITY TRADEOFF (RQ3B)

ϵ	σ	Acc.	Macro-F1	R@5	Retr.
8	1.2112	26.0%	14.7%	0.0%	×
16	0.6056	26.0%	14.6%	0.0%	×
32	0.3028	26.0%	15.3%	1.0%	×
64	0.1514	27.0%	15.7%	9.0%	✓
128	0.0757	31.0%	19.6%	57.0%	✓
256	0.0379	42.0%	30.3%	91.0%	✓
512	0.0189	37.0%	26.6%	93.0%	✓

D. RQ3: Privacy-Utility Tradeoff and Security Probes

Extended DP sweep. Figure 6 and Table IV assess accuracy and Recall@5 in the extended range of ϵ . For $\epsilon \leq 32$, Recall@5 is at most 1%, asserting that $\sigma \geq 0.30$ significantly distorts cosine similarity in the FAISS index. The retrieval-recovery threshold is set at $\epsilon=64$ ($\sigma=0.1514$, Recall@5=9.0%) with usable retrieval occurring at $\epsilon=128$ (57.0%) and near complete recovery at $\epsilon=256$ (91.0%). Accuracy limits are set to the recovery threshold, with the covariation of retrieval quality and accuracy beyond the threshold confirming that generator behaviour is strongly determined by evidence retrieved, rather than its parametric memory when the noise is at a lower level.

Security probes. Figure 7 shows two empirical evaluations of privacy from v4. The embedding inversion test assesses

Security / Privacy Probes

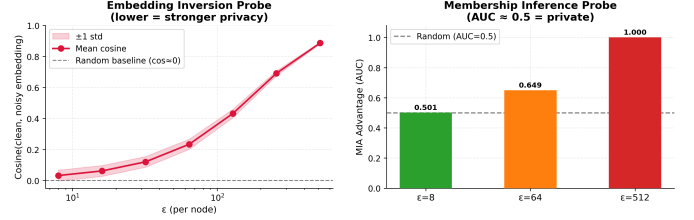


Fig. 7. Security probes indicate effective protection at $\epsilon = 8$ but significant privacy loss at $\epsilon = 512$.

Scalability Analysis

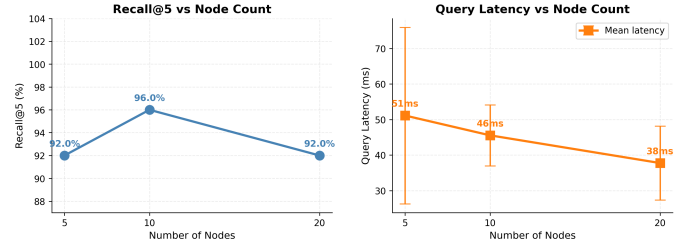


Fig. 8. Scalability results show lower latency with increasing nodes, with the best Recall@5 (96.0%) at 10 nodes.

the mean cosine similarity between the clean and the noisy embedding. At $\epsilon=8$, this measures 0.032 ± 0.035 , exhibiting a similarity that is effectively indistinguishable from random. Conversely, at $\epsilon=256$, this measure rises to 0.691 ± 0.016 , indicating high ϵ values enable retrieval but meaningfully sacrifice protection. This is confirmed by the membership inference attack (MIA) with an adversarial AUC of 0.501 (random, strong privacy) at $\epsilon=8$, 0.649 at $\epsilon=64$, and 1.000 (full leakage).

E. Scalability Analysis

Figure 8 extends the evaluation to 10 and 20 nodes, with documents re-partitioned. Recall@5 across node counts = 92.0% (5 nodes), 96.0% (10), 92.0% (20). As the number of nodes grows, mean query latency decreases from 51 ms to 38 ms, due to each sub-index per node becoming smaller and more focused. Index build time is essentially constant at ≈ 537 s across all configurations, due to per node parallelized encoding. The lack of a monotonic recall trend across node counts indicates that optimal granularity is likely more dependent on corpus size and node domain overlaps than node count.

V. CONCLUSION

FedRAG v4 indicates that privacy-preserving federated dense retrieval is feasible, with DP noise kept below a measurable threshold. A corrected BM25 is beneficial to hybrid retrieval, with the best ranking quality from Hybrid+ms-marco (Recall@5=93.5%, MRR=88.1%). Dirichlet non-IID partitioning shows no significant degradation compared to

IID federation ($p=0.528$), indicating that the approach is viable in real-world skewed hospital networks. An extended DP sweep successfully places $\varepsilon=64$ as the retrieval-recovery threshold, with security probes indicating $\varepsilon \in [64, 128]$ as a feasible privacy–utility tradeoff. Downstream QA is the main performance bottleneck. Class collapse is observed in all tested generators, with flan-t5-large attaining the best (and still low) Macro-F1 of 27.4% and no *maybe* predictions.

The focus of future work will be on instruction-tuned biomedical generators (Meditron, BioMedLM), per-sample gradient DP via Opacus, and betweeni noise retrieval that spans the range of values from $\varepsilon=64$ to $\varepsilon=256$.

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